

AML23703 – Quantum Computing

Tutorial 2: Qubits, State Vectors and Superposition Visualization

Name: Nideesh V

USN: 1GA23AI028

Department: Artificial Intelligence and Machine Learning

Semester: VII

Course Code: AML23703

Date: 30-07-2026

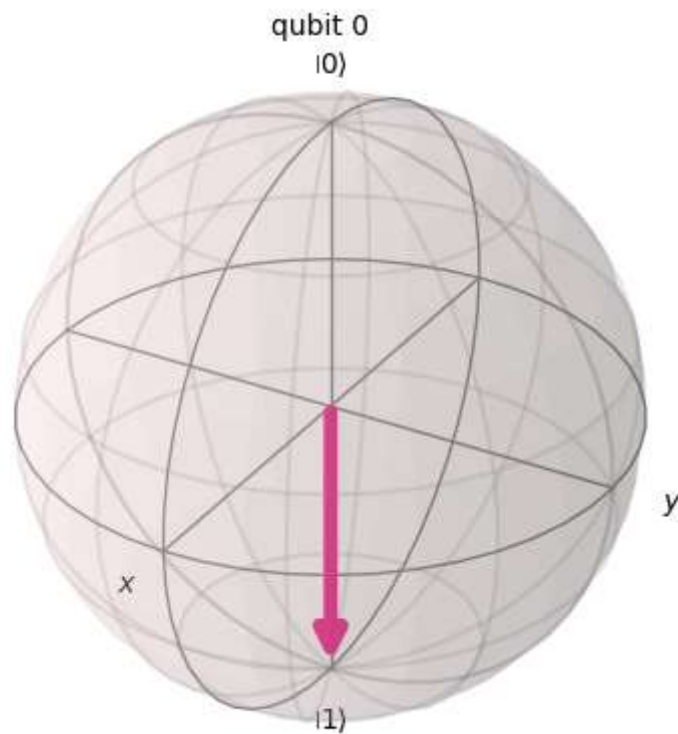
Exercises

```
In [1]: import numpy as np
        from qiskit import QuantumCircuit
        from qiskit.quantum_info import Statevector
        from qiskit.visualization import plot_bloch_multivector, plot_state_qsphere
        import matplotlib.pyplot as plt
```

Easy: $|0\rangle$ apply X gate

```
In [3]: qc=QuantumCircuit(1)
qc.x(0)
sv=Statevector.from_instruction(qc)
print(sv)
display(plot_bloch_multivector(sv))
```

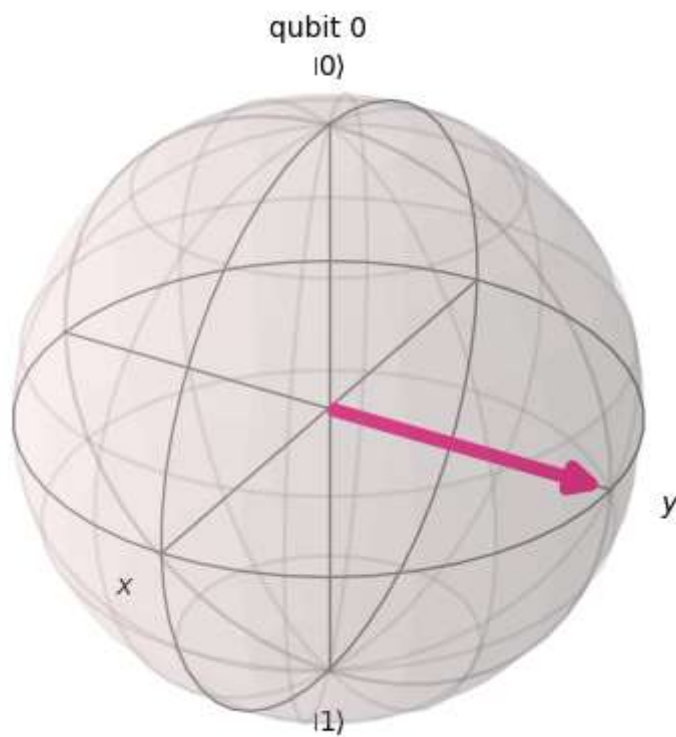
```
Statevector([0.+0.j, 1.+0.j],
            dims=(2,))
```



Medium: H then S

```
In [5]: qc=QuantumCircuit(1)
qc.h(0)
qc.s(0)
sv=Statevector.from_instruction(qc)
print(sv)
display(plot_bloch_multivector(sv))
```

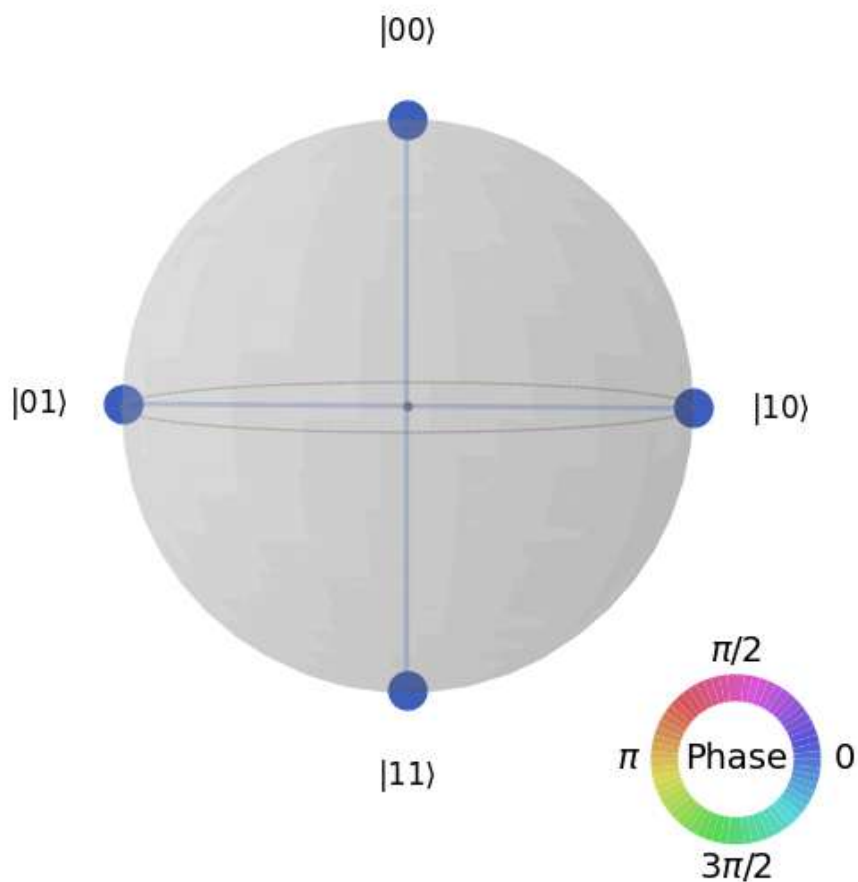
```
Statevector([0.70710678+0.j          , 0.          +0.70710678j],
            dims=(2,))
```



Hard: 2-qubit equal superposition

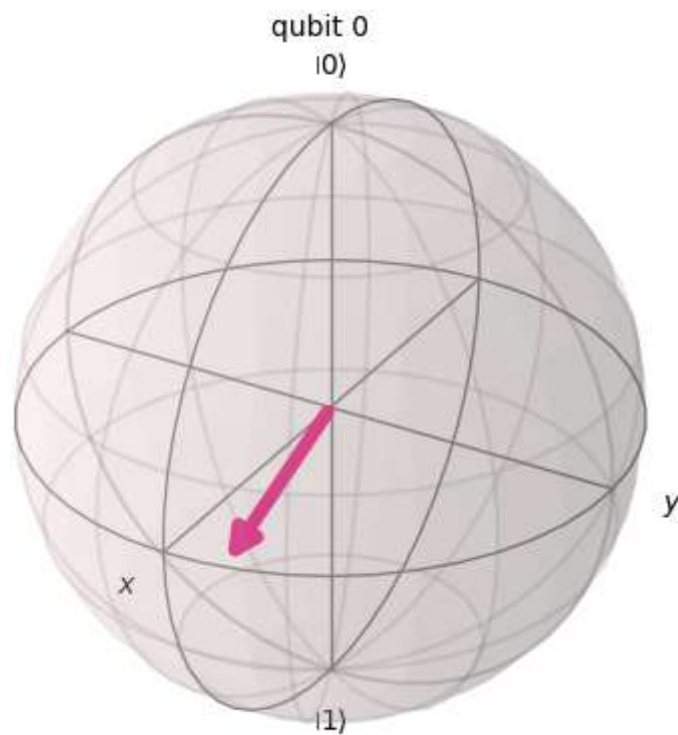
```
In [7]: qc=QuantumCircuit(2)
qc.h([0,1])
sv=Statevector.from_instruction(qc)
print(sv)
print("Normalization:",np.sum(np.abs(sv.data)**2))
display(plot_state_qsphere(sv))
```

```
Statevector([0.5+0.j, 0.5+0.j, 0.5+0.j, 0.5+0.j],
            dims=(2, 2))
Normalization: 0.9999999999999996
```



Real World: Small phase error

```
In [9]: theta=0.2  
qc=QuantumCircuit(1)  
qc.h(0)  
qc.p(theta,0)  
sv=Statevector.from_instruction(qc)  
display(plot_bloch_multivector(sv))
```



Reflection Answers

1. The sum of squared amplitudes equals 1 because total probability must always be 1.
2. The Bloch sphere represents only a single qubit. Multi-qubit entangled states require a higher-dimensional representation, so they cannot be fully visualized on a Bloch sphere.

Exported with [runcell](#) — convert notebooks to HTML or PDF anytime at [runcell.dev](#).